GAURANG PARAG BHOGLE

ELECTRICAL ENGINEERING | AI-ASSISTED DEVELOPMENT | ELECTRONICS & CREATIVE TECHNOLOGY

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PROFESSIONAL PROFILE

Electrical Engineering candidate with hands-on electronics prototyping, Arduino/C++ projects, MATLAB simulation, power and energy monitoring concepts, and industrial plant exposure. Also develops experimental AI-assisted applications, local-AI workflows and generative media pipelines. Combines technical troubleshooting, iterative testing, project documentation and digital communication skills.

TECHNICAL SKILLS

Electrical & embedded: Electrical circuits, wiring and soldering, microcontrollers, Arduino IDE, C++, electronics prototyping, inverter circuitry, energy measurement, FPV drone hardware, Betaflight, INAV, 3D printing.

Engineering, analytics & documentation: MATLAB, LaTeX, Python fundamentals, Microsoft Power BI.

AI & automation: Claude Code, Google Antigravity, ChatGPT/GPT models, Gemini, prompt-driven prototyping, debugging, Ollama, LM Studio, ComfyUI, MiniMax H3, Higgsfield.

Creative & production: Adobe Premiere Pro, After Effects, Photoshop, Canva, Filmora, video editing, motion graphics, website planning and no-code design.

ENGINEERING & ELECTRONICS PROJECTS

Smart Battery Management System (BMS) for EV | Final-year project

- Created Arduino-based control logic, simulated performance and battery management behavior in MATLAB, and prepared LaTeX documentation for an 18650-cell-based EV BMS concept focused on monitoring and safe power distribution.

IoT Smart Energy Meter | Third-year project

- Developed Arduino-based measurement and monitoring logic for voltage, current, Vrms and power consumption; simulated load scenarios in MATLAB and documented results in LaTeX.

Inverter & Obstacle Avoidance Car | Second-year project

- Designed an inverter/power-electronics circuit alongside a robotic obstacle-avoidance chassis; contributed C++ code, MATLAB simulations and LaTeX project reporting.

Music-Reactive LED Strip | First-year project

- Built a microcontroller-based audio-reactive LED visualization prototype; worked on programming, sound-to-light synchronization, simulation and documentation.

AI APPLICATIONS & GENERATIVE MEDIA PROJECTS

AI Companion App | Independent prototype

- Prototyped an interactive AI companion via AI-assisted coding, user-interface iteration, functionality testing and troubleshooting.

VTuber Streaming App | Independent prototype

- Built an experimental creator-facing VTuber streaming interface using AI-assisted development and repeated feature testing.

Job Application Automator | Independent prototype

- Explored job discovery, matching logic, LinkedIn Easy Apply interactions and browser-assisted application automation.

ComfyUI Generative Media Pipelines / AI Image & 3D Generation

- Experimented with node-based text-to-image, image-to-3D and image-to-video workflows, including MiniMax H3 product-reveal prompting on an RTX 3080-class local system.

PROFESSIONAL EXPERIENCE

Intern | Preflight Labs LLP | Aug 2026 - Present

- Contributed to video editing and marketing content, assembled and built drones, and supported 3D printing and hands-on prototyping activities.

Electrical Engineering Intern | Rashtriya Chemicals and Fertilizers (RCF) | Dec 2024 - Jan 2025

- Observed A.N. Melt Plant operations, industrial electrical equipment, maintenance practices and associated plant processes.

Technical Collaborator | Space Root Education | Dec 2024 - Jan 2025

- Supported drone component selection, Betaflight configuration and hardware troubleshooting; explained drone technical concepts to educational partners.

Freelance Content Creator & Editor | YouTube | 2020 - Present

- Produced game-review videos from scripts and voiceovers through editing, graphics, thumbnails and audience engagement using Adobe creative software.

EDUCATION

Bachelor of Engineering (B.E.), Electrical Engineering | Fr. Conceicao Rodrigues Institute of Technology, Vashi | 2022 - Present

CGPA reported in existing resume: 5.8/10 | Higher Secondary (Science): 51% | Secondary School: 66%

CERTIFICATIONS

Demystifying Generative AI for Leaders - Infosys (2024) | Python Essential Training - LinkedIn Learning (2024) | Getting Started with Power BI - LinkedIn Learning (2024) | Young IoT Prodigy - Infosys (2023) | Basics of Customer Services - TCS iON (2024) | Basics of Inventory Management - TCS iON (2024) | Cyber Security Overview - Infosys (2026) | Introduction to Cyber Security - Infosys (2026)

INTERESTS

FPV drone building and racing | Rapid prototyping and 3D printing | Home automation | Technology communication | Gaming reviews and content creation